

Uniqueness: Let, there exist two real numbers r and s such that
 $ar+b=as+b=0 \Rightarrow ar=as \Rightarrow a(r-s)=0 \because a \neq 0, \therefore (r-s)=0$
 $\Rightarrow \boxed{r=s}$. This establishes the uniqueness part of the proof.

1.8.5 Proof Strategies

FORWARD AND BACKWARD REASONING

Whichever method you choose, you need a starting point for your proof. To begin a direct proof of a conditional statement, you start with the premises. Using these premises, together with axioms and known theorems, you can construct a proof using a sequence of steps that leads to the conclusion. This type of reasoning, called Forward reasoning, is the most common type of reasoning used to prove relatively simple results. Similarly, with indirect reasoning you can start with the negation of the conclusion and, using a sequence of steps, obtain the negation of the premises.

Unfortunately, forward reasoning is often difficult to use to prove more complicated results, because the reasoning needed to reach the desired conclusion may be far from obvious. In such cases it may be helpful to use backward reasoning. To reason backward to prove a statement q , we find a statement p that we can prove with the property that $p \rightarrow q$.

Example 14: Given two positive real numbers x and y , their arithmetic mean is $(x+y)/2$ and their geometric mean is $\sqrt{xy}$. When we compare the arithmetic and geometric means of pairs of distinct positive real numbers, we find that the arithmetic mean is always greater than the geometric mean. [For example, when $x=4$ and $y=6$, we have $5=(4+6)/2 > \sqrt{4 \cdot 6} = \sqrt{24}$]

Can we prove that this inequality is always True?

Solution: Let, x and y be two distinct positive reals.
Suppose, for the sake of contradiction, $\frac{(x+y)}{2} \leq \sqrt{xy}$